

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-af89bf49162365201.jsonl`  
- SHA-256: `f68c979922c7c44aac13c1f4896046697a889c08132d0a15c40fea2b5ce28a23`  
- Source modified: `2026-06-06T17:39:30+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:38:27.937Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Scoreboard/score-overlay OCR can be used as the primary temporal-segmentation signal to detect event boundaries (wicket-taking deliveries) in racket-sport-adjacent sports video: frames are sampled, the scorecard ROI is OCR'd, and an event is declared when the recognized score count increments relative to the previous valid reading."

****Source:**** <https://arxiv.org/html/2510.18405> (primary)

****Supporting quote:**** "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:38:30.558Z)

I'll verify this claim by examining the source paper and checking the supporting quote.

3. assistant (2026-06-06T17:38:34.179Z)

Now let me fetch the source paper to verify the claim.

4. user (2026-06-06T17:38:42.003Z)

Paper Analysis: Cricket Video Analysis Using OCR and YOLOv8

(1) Sport Analyzed

The paper analyzes ****cricket****, specifically focusing on wicket-taking deliveries in broadcast match videos.

(2) Scoreboard/Score-Overlay OCR for Event Detection

Yes, the system uses OCR on scoreboard overlays to detect wicket event boundaries. The framework employs this as its primary mechanism for identifying important moments.

(3) Wicket Event Detection Methodology

The exact process is described as follows:

****Frame sampling and scorecard isolation:**** "The wicket segmentation stage begins by sampling frames from the input broadcast video at fixed temporal intervals, with a default interval of 0.1 s. For each sampled frame, a predefined region of interest (ROI) is used to isolate the scorecard area, which is typically located near the lower portion of the frame."

****Event declaration logic:**** "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

The system preprocesses the scorecard region using grayscale conversion, power transformation, morphological operations, and median blur before applying OCR.

(4) Accuracy Results for OCR-Based Event Detection

The paper does **not** report quantitative accuracy metrics for the OCR-based wicket segmentation stage. It only notes that "The OCR-guided scorecard analysis was effective in identifying wicket-taking delivery segments across multiple match videos and scorecard layouts," offering qualitative rather than numerical performance evaluation.

5. assistant (2026-06-06T17:38:45.776Z)

The quote is accurate and the methodology matches the claim. Now let me check the critical issues: (1) the sport is cricket, not racket-sport; (2) accuracy claims; (3) whether OCR-as-primary-segmentation is applicable to racket sports / disputed.

6. user (2026-06-06T17:38:54.928Z)

Web search results for query: "score overlay OCR reliability sports broadcast challenges occlusion fonts limitations"

Links: [{"title": "Scoreboard OCR Meets Logo Metrics: From Screen Time to Impact Time ? api4ai", "url": "https://api4.ai/blog/scoreboard-ocr-meets-logo-metrics-context-matters"}, {"title": "Scoreboard OCR Meets Logo Metrics: Context Matters", "url": "https://medium.com/@API4AI/scoreboard-ocr-meets-logo-metrics-context-matters-14824e6c9325"}, {"title": "Scoreboard Overlay with OCR | Spiideo Help Center", "url": "https://support.spiideo.com/en/articles/8707911-scoreboard-overlay-with-ocr"}, {"title": "Live scoreboard overlay: elevate your game and sport streaming", "url": "https://scoreleader.com/live-overlay-streaming/"}, {"title": "Best Uno Overlays Alternatives for Sports Streaming (2026)", "url": "https://keepthescore.com/blog/posts/uno-overlays-alternatives/"}, {"title": "A Simple and Effective Temporal Grounding Pipeline for Basketball Broadcast Footage", "url": "https://arxiv.org/pdf/2411.00862"}, {"title": "Automated Wicket-Taking Delivery Segmentation and Trajectory-Based Dismissal-Zone Analysis in Cricket Videos Using OCR-Guided YOLOv8", "url": "https://arxiv.org/pdf/2510.18405"}, {"title": "Building Scalable Video Understanding Benchmarks through Sports", "url": "https://arxiv.org/pdf/2301.06866"}]

Based on the search results, here's what I found regarding OCR reliability for sports broadcast score overlays and the challenges involved:

Key Challenges

Basketball broadcasters use unique game clock graphics and fonts ? making any hard-coded solution impractical. Additionally, unprocessed video corpus includes irrelevant moments (ad breaks, breaks in play) and unpredictable occlusions of key text regions.

Locating the scorecard across frames, as well as extracting match state from it is non-trivial ? due to the noisy and dynamic nature of the scorecard. More specifically, scorecards can get occluded by advertisements, move to a different position on the screen, or change their attributes (format, shape, color) during gameplay.

Addressing Quality Factors

Quality factors include size, centrality, occlusion, and sharpness. To improve reliability, venues can provide structured scoreboard data (e.g., via Sportzcast/Scorebot devices), which can be run alongside OCR for redundancy and cross-checks (clock cadence, legal score transitions, period changes).

Practical Implementation

Scoreboard OCR is an intelligent tool that automatically detects and updates scores by reading them directly from your video feed, and is especially useful for real-life sports events where manual updating would be impractical.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-06T17:38:55.503Z)

Web search results for query: "scoreboard OCR point segmentation tennis badminton broadcast video rally detection accuracy"

Links: [{"title": "(PDF) Analyzing Racket Sports From Broadcast Videos", "url": "https://www.academia.edu/70243184/Analyzing_Racket_Sports_From_Broadcast_Videos"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/html/2603.13397v2"}, {"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition | Request PDF", "url": "https://www.researchgate.net/publication/388099044_VideoBadminton_A_Video_Dataset_for_Badminton_Action_Recognition"}, {"title": "ViSTec: Video Modeling for Sports Technique Recognition and Tactical Analysis", "url": "https://arxiv.org/pdf/2402.15952"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "SmartTennisTV: Automatic indexing of tennis videos", "url": "https://arxiv.org/pdf/1801.01430"}, {"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos", "url": "https://arxiv.org/pdf/2002.08097"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}]

Based on the search results, here's what I found regarding scoreboard OCR, point segmentation, rally detection, and related techniques in tennis and badminton broadcast video analysis:

Point Segmentation Accuracy

Research on badminton video analysis achieved 95.44% point segmentation accuracy, demonstrating strong performance in identifying distinct rally segments from broadcast footage.

Scoreboard OCR and Recognition

For scoreboard understanding, researchers localize and crop scoreboard regions and apply OCR to the center frame of each rally clip for score recognition. Additionally, a score-based indexing system segments tennis matches into rallies and recognizes scores using contextual knowledge of the tennis scoring system.

Rally Detection Methods

Different approaches have been developed for rally segmentation:

- For efficient processing across hundreds of hours of video, researchers employ a lightweight, purely audio-based detection model trained to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track.
- An alternative method leverages the observation that in broadcast tennis videos the camera is only overhead when the rally is in play, using HOG features and a γ -squared kernel SVM to label frames as rally or non-rally frames.

Detection Models

For detecting the start of service, researchers have used Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN) models while gradient computation determined the

end of the rally.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-06T17:39:08.287Z)

I have enough to assess. Let me verify the key facts of the source paper once more ? that it does NOT report quantitative accuracy and the sport is cricket, not a racket sport.

The findings are now clear. Let me synthesize my adversarial assessment.

The claim states: "Scoreboard/score-overlay OCR can be used as the primary temporal-segmentation signal to detect event boundaries... frames are sampled, the scorecard ROI is OCR'd, and an event is declared when the recognized score count increments relative to the previous valid reading."

Key findings:

1. **The quote is accurately quoted and faithfully describes the paper's method.** The paper (2510.18405) does exactly this: samples frames at 0.1s, isolates a scorecard ROI, OCRs it, and declares a wicket event when the count increments vs. the preceding valid reading. No misquote, no overreach on the mechanism itself.
2. **Methodology is corroborated independently** ? score-overlay OCR for rally/point segmentation is a well-established technique (SmartTennisTV uses score-based indexing; tennis/badminton work crops scoreboard regions and OCRs them; basketball temporal-grounding uses game-clock OCR). So the technique is real and used in racket sports too.
3. **Caveats that the claim itself acknowledges** ("racket-sport-adjacent", cricket is the actual sport). The claim is honestly hedged about the sport.
4. **The source does NOT report quantitative accuracy** ? only qualitative ("effective... across multiple match videos"). But the claim does not assert any accuracy figure; it only describes a method ("can be used as..."). So the lack of metrics does not refute the (descriptive, capability-level) claim.
5. **Limitations exist** (occlusion, font/layout changes, scorecard not always present) but these qualify reliability, not the existence/validity of the method as a usable signal.

The claim is descriptive and capability-level ("can be used as the primary signal"), accurately supported by the quote, corroborated by multiple independent sources, current (2025 paper), and from a primary arXiv source. It is not refuted.

9. user (2026-06-06T17:39:19.313Z)

Structured output provided successfully

10. assistant (2026-06-06T17:39:30.492Z)

I've completed the adversarial verification. The claim is **not refuted** (confidence: high).

The supporting quote is verbatim and the methodology in the source paper (arXiv 2510.18405) matches the claim exactly ? frame sampling at 0.1s, scorecard ROI isolation, OCR, and event declaration on count increment vs. the preceding valid reading. All four refutation angles failed: no misquote, the sport mismatch is honestly hedged ("racket-sport-adjacent") and the technique is independently used in actual racket sports, the missing accuracy metrics don't contradict a capability-level claim, and the source is a current primary arXiv paper. The only real weakness ? OCR brittleness under occlusion/font/layout changes ? qualifies reliability but does not refute the method's validity.